

OS Core components and System calls

Operating System (kernel)
manages resources and provides services to applications.

System call Interface

Applications (user programs) request services via system calls (API)

Common system calls

- Process control (fork, exec, exit, wait)
- memory management (mmap, brk, munmap)
- communication (pipe, socket, shm)

Requests services

Process management

manages process and their execution

- process creation
- CPU scheduling
- context switching
- synchronization

File system

manages file and directories

⇒ Interacts with process management, memory management, I/O system.

Memory management

manages main memory resources

- Allocation / deallocation
- Paging / Segmentation
- virtual memory
- Protection & Sharing

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System calls Route to appropriate module.

I/O Devices

- manages I/O devices
- Device drivers
- Interrupt handling
- I/O scheduling
- Buffering / caching

Hardware Resources

- CPU
- Main memory
- Storage devices
- I/O devices

Comparison of operating system types.

operating system (General)

Different os types designed to handle work load in different ways

Batch os

How it works

jobs with similar needs are collected

Key features

- job scheduling
- spooling
- NO direct user interaction

Advantages

- High throughput
- Efficient for large volume jobs

Limitations

- Simple Limitation
- Long turnaround time
- Poor for urgent jobs

Best suited for

- Pay roll Systems
- Billing Systems
- Scientific computations

Time sharing os

cpu time divided into small time slices and allocated to mult users

- Multi programming
- CPU scheduling
- interactive respons

- Good cpu utilization
- interactive and responsive
- supports multiple users

- complex scheduling
- data protection issues
- context switching overhead

- Multi user systems
- education, offices

Distributed os

Manages a group of independent computers and makes them appear

- Resource sharing
- communication between nodes
- Transparency

- Load sharing
- Scalability
- fault tolerance

- complex to design
- Network dependency

- Cloud computing
- Large scale enterprises

Real time os

provides guaranteed responses within strict time constraints

- Deterministic behaviour
- Priority based preemption
- Predictable performance

- Limited flexibility
- complex programming

- Embedded systems
- Industrial control
- Avionics, Medical devices